

SOFTWARE TEST PLAN (STP)

Project: A Snake Game

A Modular Snake Game with Multiple Game Modes, Power-Ups, Survival Challenges, Multiplayer and Leaderboards

Version: 1.0 Course: Software Engineering Date: 16-09-2026 Team 6 Status: Draft

Authors

Name	SRN	Primary responsibility
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Revision History

Version	Date	Author	Change Summary
1.0	16-09-2026	Team 6	Initial test plan derived from SRS v0.1 and SAD v1.0

1. Introduction

Purpose: This Software Test Plan (STP) defines the testing objectives, scope, strategy, resources, schedule, responsibilities, risks, and traceability approach for A Snake Game. It is based on the A Snake Game SRS v0.1 and the Software Architecture and Design Specification v1.0.

Scope: Testing covers player registration and authentication, profiles, logout, game initialization and controls, five game modes (Classic, Obstacle, Power-Up, Two-Player Battle and Survival), movement, food, collision detection, scoring, score persistence, leaderboard, administration/security controls, performance, usability, maintainability and compatibility. Physical gaming hardware, commercial payment systems and external gaming services are outside the SRS scope.

References: A Snake Game — Software Requirements Specification v0.1 (09-09-2026); Software Architecture and Design Specification v1.0 (09-09-2026); Requirements Traceability Matrix contained in the SRS.

Definitions: SRS – Software Requirements Specification; STP – Software Test Plan; RTM – Requirements Traceability Matrix; UI – User Interface; FR – Functional Requirement; NFR – Non-Functional Requirement; FPS – Frames Per Second; Admin – System Administrator; Game Session – a single instance of gameplay; Power-Up – temporary gameplay ability/effect; Leaderboard – ranked list of player scores.

2. Test Items

- User Management / Authentication Module
- User Interface Module
- Game Engine and Snake Controller
- Game Mode Manager and individual game modes
- Collision Detection Module
- Food Manager
- Power-Up Manager
- Difficulty Manager
- Score Manager
- Leaderboard Module

- Database Layer / Database Module
- Administration Module
- Statistics Module

3. Features to be Tested

Functional requirements:

Req ID	Feature	Test Case
SNAKE-F-001	Player Registration	TC-AUTH-001
SNAKE-F-002	Player Authentication	TC-AUTH-002
SNAKE-F-003	Player Profile	TC-PROFILE-001
SNAKE-F-004	Logout	TC-AUTH-003
SNAKE-F-005	Start Game	TC-GAME-001
SNAKE-F-006	Game Mode Selection	TC-MODE-001
SNAKE-F-007	Difficulty Selection	TC-DIFF-001
SNAKE-F-008	Pause and Resume	TC-GAME-002
SNAKE-F-009	Snake Movement	TC-MOVE-001
SNAKE-F-010	Food Consumption	TC-FOOD-001
SNAKE-F-011	Food Generation	TC-FOOD-002
SNAKE-F-012	Collision Detection	TC-COLLISION-001
SNAKE-F-013	Score Calculation	TC-SCORE-001
SNAKE-F-014	Obstacle Mode	TC-OBSTACLE-001
SNAKE-F-015	Power-Up Mode	TC-POWERUP-001
SNAKE-F-016	Two-Player Battle	TC-MULTI-001
SNAKE-F-017	Survival Mode	TC-SURVIVAL-001
SNAKE-F-018	Classic Mode	TC-CLASSIC-001
SNAKE-F-019	Save Game Score	TC-SCORE-002
SNAKE-F-020	Leaderboard	TC-LEADER-001

Non-functional requirements:

Req ID	Requirement / Target	Test Case
SNAKE-NF-001	Valid movement input processed within 100 ms	TC-NF-PERF-001
SNAKE-NF-002	At least 30 FPS during normal single-player gameplay	TC-NF-PERF-002
SNAKE-NF-003	Stored scores preserved without unintended loss/modification	TC-NF-REL-001
SNAKE-NF-004	New player starts Classic Mode within three primary menu interactions after login	TC-NF-USE-001
SNAKE-NF-005	Major components are modular for independent testing and maintenance	TC-NF-MAIN-001
SNAKE-NF-006	Consistent gameplay behavior across supported operating environment	TC-NF-COMP-001

Security requirements:

Req ID	Security requirement	Test Case
SNAKE-SR-001	Passwords use secure one-way hashing; plaintext passwords are not stored	TC-SEC-001
SNAKE-SR-002	Administrator functions restricted to authenticated and authorized administrator accounts	TC-SEC-002
SNAKE-SR-003	User-provided input validated before processing or storing	TC-SEC-003

Req ID	Security requirement	Test Case
SNAKE-SR-004	Normal player interfaces cannot directly modify stored scores/leaderboard records	TC-SEC-004
SNAKE-SR-005	Authenticated sessions terminate according to session-management policy and protected functionality is blocked after logout	TC-SEC-005

4. Features Not to be Tested

- Physical gaming hardware, because the SRS defines the project as a software implementation.
- Commercial payment systems.
- External gaming services.
- Future enhancements not included in the current scope, such as online multiplayer, AI-controlled snakes, custom map creation, tournament mode, cloud synchronization, achievements/badges, replay system and mobile version.
- Implementation-specific tools or technologies that are marked TBD in the SAD/SRS.

5. Test Approach / Strategy

Levels: Unit testing at module level; integration testing between UI, game modules, score/database and authentication/security components; system testing of end-to-end gameplay; acceptance testing against the SRS acceptance criteria.

Types: Functional testing; regression testing after fixes/changes; performance testing for ≤ 100 ms input response and ≥ 30 FPS; usability testing for menu navigation; compatibility testing on supported Windows configurations; security validation for authentication, authorization, input validation, credential handling and score/leaderboard integrity; reliability testing for score persistence.

Entry Criteria: A testable build is available; required game modules are integrated sufficiently for the planned tests; supported Windows test environment is ready; test data/accounts are available; defects that prevent basic execution have been resolved or documented.

Exit Criteria: All planned test cases are executed; all high-priority functional requirements are verified; required non-functional and security targets are met; no critical defects remain in the final build; acceptance criteria in the SRS are satisfied.

5.1 Security Validation

- Verify one-way password hashing and absence of plaintext password storage.
- Verify administrator functions require authenticated and authorized administrator accounts.
- Verify user input is validated before processing or storage.
- Verify normal player interfaces cannot directly modify stored scores or leaderboard records.
- Verify logout/session termination prevents access to protected functionality according to configured policy.

6. Test Environment

Hardware: Windows-based desktop/laptop system; keyboard for player input; display/monitor with resolution of at least 1280×720 .

Software: A Snake Game test build; supported modern runtime/browser environment depending on implementation technology; configured local/application database.

Tools: Testing/automation and defect-tracking tools: TBD, because the SRS/SAD do not specify final tools.

Test Data: New and existing player accounts, valid/invalid credentials, permitted profile data, gameplay sessions, multiple scores for leaderboard verification, administrator account, invalid input values, and representative gameplay states.

7. Test Schedule

- Test case design: TBD

- Test environment setup: TBD
- Test execution start: TBD
- Test execution end: TBD
- Regression testing: After defect fixes and before final acceptance
- Acceptance testing: TBD

8. Test Deliverables

- Software Test Plan
- Test cases and test data
- Test scripts, if automation is used
- Test execution logs/results
- Defect reports
- Requirements Traceability Matrix / coverage record
- Test metrics and daily execution status
- Final Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA/Test Lead	Vishwas Hosamath	Prepare test plan, design/execute tests, security validation, defect reporting, traceability and test summary.
Development Lead	Thejaswi S	Support test builds, core game engine fixes, defect triage and technical clarification.
Multiplayer/Communication	Vishnu Maddirala	Support Two-Player Battle testing, input handling and multiplayer/communication interfaces.
UI/Game Modes	V S Shashidharan	Support UI, visualization, game-mode behavior and usability testing.
Course/Project Evaluator	Course Faculty	Review correctness, completeness and acceptance of project requirements/tests.

10. Risks and Mitigation

Risk	Mitigation
Unstable test build or late build delivery	Request early test builds and prioritize smoke testing before full execution.
Database/account data unavailable	Prepare representative test accounts/data before execution and verify database availability.
Inconsistent gameplay across supported environments	Perform compatibility testing and verify consistent game-state transitions.
Unauthorized score/leaderboard modification	Execute security/integrity tests and verify player interfaces cannot directly modify stored records.
Authentication/session defects block protected functions	Run authentication/security smoke tests early and retest after fixes.
Performance targets not met	Measure input response time and FPS during normal gameplay and isolate affected modules.

11. Assumptions & Dependencies

- The supported Windows environment and required runtime are available for testing.
- A configured local/application database is available where required.

- Test accounts and representative game data can be created.
- The final implementation will provide the modules and interfaces described by the SRS/SAD.
- Final testing tools remain subject to the implementation and course/project setup because the SAD lists tools as TBD.

12. Suspension & Resumption Criteria

- Suspend testing if the test environment becomes unavailable or unstable enough to prevent reliable execution.
- Suspend testing if the build has blocking defects that prevent execution of a substantial portion of planned test cases.
- Suspend testing if a critical dependency required for authentication, database access or core gameplay is unavailable.
- Resume testing when blocking defects are resolved, the environment is stabilized, and a testable build is available.

13. Test Case Management & Traceability

- The RTM in the SRS maps each functional requirement to a corresponding test case. This test plan extends that mapping to non-functional and security requirements so that every requirement has a planned verification activity.
- Requirement coverage is reviewed before final sign-off. Test results are recorded against the corresponding test case IDs, and failed cases are linked to defect reports where applicable.

14. Test Metrics & Reporting

- % of planned test cases executed
- % of test cases passed/failed/blocked
- Defect count and severity
- Defect density and defect aging
- Requirement coverage
- Performance measurements: movement response time and FPS
- Number of critical/open defects at exit
- Security test pass status
- Daily execution status and final Test Summary Report

15. Approvals

- QA/Test Lead: Vishwas Hosamath — Signature/Date: _____
- Development Lead: Thejaswi S — Signature/Date: _____
- Team Representative / Multiplayer: Vishnu Maddirala — Signature/Date: _____
- UI/Game Modes Representative: V S Shashidharan — Signature/Date: _____
- Course Coordinator: _____ — Signature/Date: _____

Appendix A — Detailed Test Cases

Test cases are derived from the SRS functional, non-functional and security requirements.

ID	Test Item	Test Procedure	Expected Result
TC-AUTH-001	Player Registration	Enter a new unique username and password and submit registration.	Account is created successfully; duplicate/invalid input is rejected.
TC-AUTH-002	Player Authentication	Enter valid and invalid registered credentials.	Valid credentials authenticate the player; invalid credentials are rejected.
TC-AUTH-003	Logout	Log in and select Logout, then attempt to access protected functions.	Session ends and protected functionality is inaccessible.
TC-PROFILE-001	Player Profile	View and update permitted profile information.	Permitted information is displayed and valid updates are saved.
TC-GAME-001	Start Game	Log in, select a valid game mode and difficulty, and start.	A new game session starts with the selected configuration.
TC-MODE-001	Game Mode Selection	Open mode selection and choose each of the five modes.	Classic, Obstacle, Power-Up, Two-Player Battle and Survival can be selected.
TC-DIFF-001	Difficulty Selection	Select each available difficulty before starting a game.	Selected difficulty is accepted and applied to gameplay.
TC-GAME-002	Pause and Resume	Pause an active game and then resume it.	Gameplay stops while paused and continues correctly after resume.
TC-MOVE-001	Snake Movement	Use keyboard controls during gameplay, including direction changes.	Snake moves continuously according to input and configured speed.
TC-FOOD-001	Food Consumption	Move the snake onto generated food.	Food is consumed, snake length increases and score increases according to rules.
TC-FOOD-002	Food Generation	Start/restart games and observe generated food positions.	Food appears at valid unoccupied board positions.
TC-COLLISION-001	Collision Detection	Trigger wall, self, obstacle and applicable opponent-snake collisions.	Applicable collision is detected and the game state changes correctly.
TC-SCORE-001	Score Calculation	Perform scoring gameplay events across applicable modes.	Score is calculated according to the defined game rules.
TC-OBSTACLE-001	Obstacle Mode	Start Obstacle Mode and collide/avoid obstacles.	Obstacles appear and collision with an obstacle results in Game Over.
TC-POWERUP-001	Power-Up Mode	Collect configured power-ups and observe their effects and expiration.	Power-up effects are applied for configured durations and expire correctly.
TC-MULTI-001	Two-Player Battle	Start Two-Player Battle and control both snakes using separate controls.	Both players control separate snakes simultaneously on the same board; collisions and scoring work.
TC-SURVIVAL-001	Survival Mode	Play Survival Mode for increasing durations.	Difficulty progressively increases through configured changes such as speed, obstacles or reduced safe area.
TC-CLASSIC-001	Classic Mode	Play a single-player Classic Mode game and collect food.	Snake grows and score increases; applicable invalid collision ends the game.
TC-SCORE-002	Save Game Score	End an authenticated player's game and inspect saved game data.	Final score and relevant game information are stored.
TC-LEADER-001	Leaderboard	Complete games with multiple scores and open leaderboard.	Leaderboard displays player scores in ranked order.
TC-NF-PERF-001	Movement Response Time	Measure time between valid keyboard input and corresponding movement response.	Measured response time is ≤ 100 ms under normal operating conditions.
TC-NF-PERF-002	Frame Rate	Measure FPS during normal single-player gameplay.	Measured FPS is ≥ 30 on supported hardware.
TC-NF-REL-001	Score Persistence	Store a score, retrieve it repeatedly and compare results.	Retrieved score matches stored score without unintended modification/loss.
TC-NF-USE-001	Classic Mode Navigation	After successful login, count primary menu interactions needed to start Classic Mode.	A new player can start Classic Mode within three primary menu interactions.

ID	Test Item	Test Procedure	Expected Result
TC-NF-MAIN-001	Modular Architecture Review	Review architecture/code organization against defined modules.	Major components are separated to support independent testing and maintenance.
TC-NF-COMP-001	Compatibility	Run gameplay and core functions across supported Windows configurations.	Gameplay behavior is consistent across supported environments.
TC-SEC-001	Password Storage	Inspect account storage/credential handling in the test environment.	Passwords are stored using one-way hashing and no plaintext password is stored.
TC-SEC-002	Admin Authorization	Attempt administrator functions with player and authorized admin accounts.	Only authenticated and authorized administrator accounts can access admin functions.
TC-SEC-003	Input Validation	Submit valid, malformed and unexpected user-provided inputs.	Invalid input is rejected or safely handled before processing/storage.
TC-SEC-004	Score/Leaderboard Integrity	Attempt to alter stored scores or leaderboard data through normal player interfaces.	Player interfaces do not permit direct modification of stored records.
TC-SEC-005	Session Termination	Log out and attempt to reuse protected functionality/session state.	Protected functionality is inaccessible after logout according to configured policy.

Appendix B — Traceability Summary

SNAKE-F-001 through SNAKE-F-020 are mapped one-to-one to the test case IDs specified in the SRS. The STP adds verification cases for SNAKE-NF-001 through SNAKE-NF-006 and SNAKE-SR-001 through SNAKE-SR-005.